



MUSYNERGY: A framework for music collaboration discovery based on neural networks and graph analysis

Alejandro Fernandez-Sanchez, Pedro J. Navarro, Fernando Terroso-Saenz *

Technical University of Cartagena (UPCT), Plaza Cronista Isidoro Valverde, Cartagena, 30202, Spain

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ABSTRACT

The music industry has been reshaped by the rise of artist collaborations, driven by digital technologies, streaming platforms, and the globalization of music. While existing research has examined the cultural and commercial impact of collaborations, few efforts have focused on recommendation systems to assist musicians in discovering potential creative partners. Moreover, most approaches rely on proprietary data, limiting scalability and reproducibility. This paper presents MUSYNERGY, a novel framework for music collaboration discovery based on neural networks and graph analysis. MUSYNERGY builds a Heterogeneous Knowledge Graph (HKG) using open data from MusicBrainz, representing relationships among artists, tracks, and musical attributes over five decades. By formulating collaboration discovery as a link prediction task, the system identifies new, plausible collaborations between artists with no prior joint work. This open, scalable framework addresses current limitations in data accessibility and supports innovation, transparency, and cultural exchange in the global music landscape through data-driven collaboration discovery.

1. Introduction

The evolution of the music industry over the past 20 years has been significantly influenced by a surge in collaborations among artists, reshaping how music is produced, marketed and consumed. This trend has been propelled by advancements in digital technology, the rise of streaming platforms, and the globalization of music, which have all contributed to a more interconnected and genre-fluid musical landscape. Collaborations have emerged as a vital strategy for artists seeking to enhance their visibility, diversify their sound, and engage with broader audiences, turning partnerships into a defining feature of contemporary music culture [1,2].

Notable examples of successful collaborations include hits like *Uptown Funk* by Mark Ronson featuring Bruno Mars and *I Don't Care* by Ed Sheeran and Justin Bieber, which demonstrate the commercial success and creative potential of these partnerships. The fusion of various musical styles and genres has led to innovative soundscapes that resonate with diverse listener demographics, pushing the boundaries of traditional musical norms [3]. Furthermore, collaborations often serve as gateways to new creative realms, allowing artists to experiment and grow while sharing their distinct influences and experiences [4]. As the music industry continues to adapt to technological advancements and evolving consumer behaviors, the role of collaborations remains central to fostering innovation, enhancing cultural exchange and connecting artists with diverse audiences worldwide [5].

In the digital era, access to music has expanded significantly, allowing listeners to enjoy a wide range of genres and artists through various streaming platforms such as Spotify, YouTube, or Apple Music. These platforms serve as valuable data sources, fostering a growing research field within the data science community. Many studies in this area have adopted a user-centered approach, focusing on solutions for the general audiences, such as music recommendation systems [6,7] or sentiment analysis [8]. With regard to the study of musical collaborations, research efforts typically follow two main directions: (1) exploratory analyses of key factors that have influenced past collaborations among musicians [5,9] and (2) hit-song prediction models that incorporate collaboration-related features [10].

However, there is a lack of proposals focusing on the development of recommendation systems to assist musicians in identifying other artists with compatible styles or genres, potentially fostering new collaborations. In this regard, the few existing approaches in this direction rely on datasets extracted from private stakeholders, such as the aforementioned music streaming services [11]. This significantly limits the size and diversity of the available data due to the usage constraints imposed by these data sources. Consequently, there is a notable lack of studies exploring the use of open musical data to develop innovative recommendation systems aimed at fostering effective collaborations among musicians.

* Corresponding author.

E-mail address: fernando.terroso@upct.es (F. Terroso-Saenz).

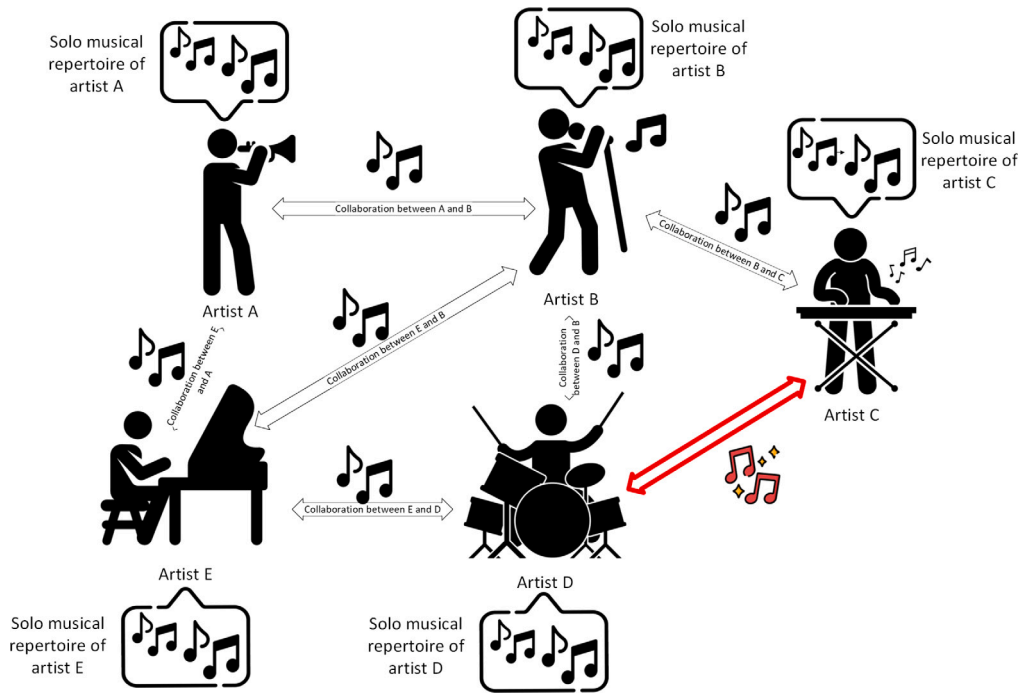


Fig. 1. Overview of the link-prediction approach followed by MUSYNERGY. The system takes as input both the collaborations and repertoire of solo songs of musicians and recommends novel and unseen collaborations among them (like the one between Artist C and Artist D in the image).

Taking into account the limitations mentioned above, this study presents *MUSYNERGY*, a framework for music collaboration discovery based on Neural Networks and Graph analysis. *MUSYNERGY* is an innovative framework that identifies potential music collaborations using neural network techniques and graph analysis. The system employs a graph-based strategy by integrating a Heterogeneous Knowledge Graph (HKG) that encapsulates the creative connections among artists, their songs, and key musical attributes. Utilizing this comprehensive dataset, *MUSYNERGY* frames the recommendation of new collaborative opportunities among artists who have not previously partnered as a link prediction problem as Fig. 1 depicts.

To achieve this, we build this HKG with MusicBrainz (MB), a community-driven open music encyclopedia that collects and provides detailed metadata on artists, tracks and releases [12]. In this way, our approach spans more than 50 years of artist collaborations.

The rest of this paper is organized as follows. Section 2 reviews current trends and techniques used in analyzing music streaming data. In Section 3, we describe the MusicBrainz dataset used in the present study. Section 4 explains the internal design of *MUSYNERGY*, while Section 5 evaluates the performance of the proposed recommender system. Next, Section 6 presents some illustrative examples of how *MUSYNERGY* can be used. Finally, Section 7 concludes with the main findings and outlines directions for future research.

2. Related work

Analyzing streaming music data has been a central focus in various research efforts within the realm of music recommendation systems [13]. In this context, a key approach has been the design of specialized recommendation systems that propose new songs based on user preferences or contextual factors. In this regard, deep learning techniques stand out for their ability to identify latent features in musical items, either independently or in a sequential fashion [14]. As a result, graph neural networks (GNNs) have been introduced to study hypergraphs that capture intricate relationships between listeners and songs [15]. A related GNN-based approach incorporating an attention mechanism was presented in [16].

Concerning other types of neural networks, the work described in [17] presents a music recommendation system using a Convolutional Recurrent Neural Network (CRNN) combined with content-based filtering. The RNN component processes sequential emotional data extracted from facial expressions, classifying emotions into seven categories. Based on this classification, the system suggests personalized music via Spotify's API, aligning song recommendations with the user's emotional state. Moreover, the work put forward in [18] explores music genre recommendation using a Multi-Layer Perceptron (MLP) trained on user questionnaire data.

Regarding Convolutional Neural Networks (CNNs), the personalized music recommendation system put forward in [7] makes use of a CNN to analyze musical data and extract complex audio features from songs, enhancing recommendation accuracy compared to traditional methods. A similar approach based on a Compact Convolutional Transformer (CCT) is stated in [19]. Moreover, the work in [20] presents a facial expression-based music recommendation system in which a CNN is employed to analyze facial emotion data extracted from images and classify emotions. These detected emotions are then used to query a song database to provide personalized music recommendations.

All the previously mentioned studies primarily aim at delivering recommendations to end users. From the artists' perspective, a significant research area in the music industry has been *Hit Song Science* [21], which seeks to anticipate a song's popularity before its official release. For instance, the study in [22] predicts song popularity using Spotify and YouTube data, including audio features, lyrics, and artist information, by applying various machine learning algorithms such as Ridge Regression or Elastic Net. Similarly, the research described in [23] predicts the success of music playlists using songs' analytical features such as danceability, energy or loudness. MLPs and Recurrent Neural Networks (RNNs) are used to classify playlists by popularity. Finally, an interesting proposal described in [24] fuses the audio characteristics of the target songs and the emotion analysis of online social networks to anticipate the popularity of the tracks.

Concerning recommendation systems for suggesting collaborations among artists, the work presented in [11] introduces a GNN-based approach to predict musical collaborations, similar to *MUSYNERGY*.

Table 1

Comparison between different approaches for music data analysis. Acronyms: BN - Binary Classifier, CNN- Convolutional Neural Network, CRNN - Convolutional Recurrent Neural Network, DNN - Deep Neural Networks, EDA - Exploratory Data Analysis, FPM - Frequent Pattern Mining, GAN - Graph Attention Network, GNN- Graph Neural Networks, LR - Linear Regression, MLP - Multilayer Perceptron, LSTM - Long Short Term Memory, RFM - Random Forest Classifier, RT - Regression Tree.

Ref.	Music dataset(s)	Feed category	Data available?	Goal	Techniques
[15]	Million Song Dataset	Open	No	Music recommendation	GNN
[16]	30Music Dataset	Open	Yes	Music recommendation	GAN
[17]	Spotify API	Proprietary	Yes	Music recommendation	CRNN
[18]	Users questionnaire	Proprietary	No	Music recommendation	RFC, MLP
[7]	Ad-hoc music files	Proprietary	Yes (upon request)	Music recommendation	CNN
[19]	GTZAN dataset	Open	Yes	Music recommendation	CCT
[20]	Spotify API	Proprietary	Yes	Music recommendation	CNN
[22]	Spotify API	Proprietary	Yes	Hit-song prediction	LR
[23]	Million Song Dataset	Open	No	Hit-song prediction	DNN, LSTM
[24]	Spotify API	Proprietary	Yes	Hit-song prediction	RFC, RT
[11]	Spotify API, LFM-2G	Proprietary	Yes	Music collaboration prediction	GNN
[25]	Billboard Hot 100, Spotify API	Open/Proprietary	Yes	Music collaboration EDA	BC
[9]	Songtree	Public	No	Music collaboration EDA	FPM
MUSYNERGY	MusicBrainz	Public	Yes	Music collaboration prediction	GNN

However, there are two key differences between such a work and MUSYNERGY. First, it relied on proprietary datafeeds like Spotify and Last.fm to compose the system whereas our proposal follows an open-data approach that allows to collect a more ambitious set of artists and tracks. Besides, its GNN architecture splits the temporal and static features of the tracks in two parallel pipelines. On the contrary, MUSYNERGY provides a lightweight architecture that combines all features into a single pipeline as described in Section 4.4. For the sake of completeness, the authors in [25] describe how the enrichment of hit-song predictors with the collaboration features of the songs produce better results. Lastly, a study focused on identifying the main factors that guide the composition of collaborative songs [9].

For the sake of completeness, Table 1 summarizes the reviewed contributions based on their datasources, scope and applied techniques.

Last, our approach aims to introduce an innovative recommendation system tailored for music artists. In contrast to previous studies, MUSYNERGY is specifically designed to suggest potential collaborations between musicians for creating new songs solely relying on open-data sources. In this sense, our method serves as an initial step before applying hit-song prediction models. Moreover, it extends beyond the exploratory research on musical collaborations discussed earlier.

3. The MusicBrainz Database

The dataset used to make up MUSYNERGY has been extracted from the *MusicBrainz Database*¹ (MBDB). It is an extensive, open music metadata repository built on the PostgreSQL relational database engine. It comprises detailed *core data*—including information on artists, release groups, releases, recordings, works, and labels—as well as numerous relationships that link these entities, together with a full history of all community edits. The core data is released under the CC0 license (effectively placing it in the public domain). Moreover, the database is updated bi-weekly through complete snapshots, ensuring that users have access to current and comprehensive music metadata for a wide range of applications.

For this study, we identified 3 different high-level entities from the MBDB, namely *artists* to collect the characteristics related to the musicians, *tracks* to gather information about their songs and *tag* which stores additional meta-data about musicians like their associated music genre. We extracted different features related to these entities and their relationships. In that sense, Table 2 summarizes the features collected for these entities and relationships along with the MBDB tables used for that purpose. As a result, a dataset comprising 1,489,250 artists, 24,324,100 tracks and 1,231,526 artistic collaborations. All

these tables, not only enable the identification of artists participating in collaborations but also capture the characteristics of these collaborations.

4. Development of MUSYNERGY

In this section, we describe the workflow followed to develop MUSYNERGY. As shown in Fig. 2, the process consisted of four key steps. First, we conducted an exploratory analysis of the *MusicBrainz Database* (Section 4.1). Based on this analysis, we constructed the HKG to represent the relevant connections and features of the various actors involved in the recommendation process (Section 4.2). Next, we designed the architecture of the GNN to be used as a recommender system (Section 4.4), trained it, and finally deployed it. The resulting system takes as input a pair of artists and a time horizon, and returns whether the two artists are likely to collaborate within the specified period.

4.1. Exploratory data analysis

To construct a heterogeneous graph that effectively represents the relationships among the key entities in the recommendation process (artists, songs and tags) we conducted an exploratory analysis of the given dataset. The key insights derived from this analysis are presented below.

To begin with, Fig. 3 presents key findings about the collaborative tracks registered in the MBDB. Specifically, Fig. 3a illustrates the number of collaborative tracks recorded over the last 10 years in the dataset. As observed, there has been a steady increase in this type of song from 2014 to 2021, followed by a slight decrease in 2022 and 2023. This trend aligns with previous studies reporting a rise in such collaborations in the global music market [25].

Furthermore, Fig. 3b shows the ratio of collaborative songs to the total musical repertoire of artists. The data reveal that this ratio is approximately 0.3 for mature artists born between 1950 and 1990. However, the ratio declines for younger artists born after 2000. This may suggest that musicians become more open to collaborating with other artists as their careers become more established, whereas emerging artists tend to be more reluctant to engage in such collaborations early in their careers.

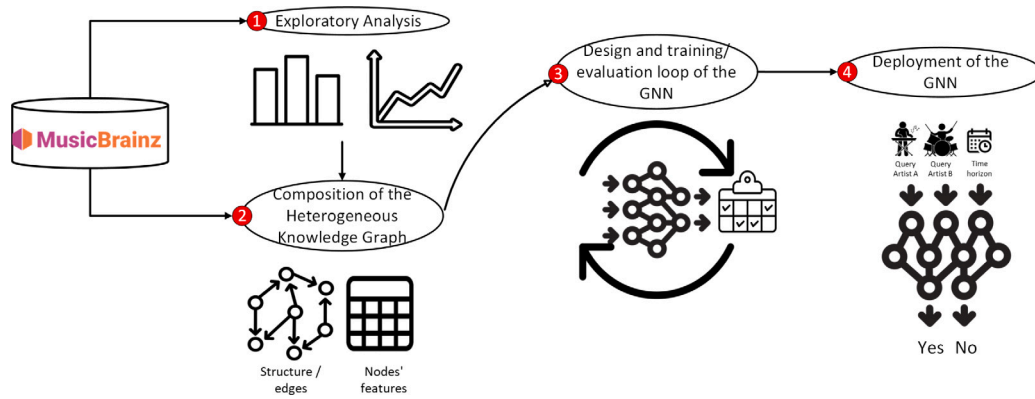
Similarly, Fig. 3c presents this ratio for music bands. Two key patterns emerge from the data. First, the proportion of collaborative songs by music bands is significantly lower than that of individual artists. Second, mature bands formed between 1950 and 1970 tend to have a higher rate of collaborations compared to younger bands, a pattern consistent with the one observed among individual artists.

¹ https://musicbrainz.org/doc/MusicBrainz_Database.

Table 2

Entities, relationships and their features extracted from the indicated MBDB tables. The last three entities are different relationships among the entities. Note that Artist-Artist, Artist-Track and Artist-Tag rows report relationships between the entities.

Entity/Rel.	Features	Description	MBDB tables
Artist	IDs	MB unique identifier	artist, l_artist_artist, link, link_type
	Names	Name of the artist	
	Type	Type of artist (person, group, character, orchestra, choir, other)	
	Gender	Gender of artist (male, female, not applicable, non-binary, other)	
	Ended	Probability that their career is over (decease, dissolved...)	
	Begin dates	Start year (birth, band creation...)	
	End dates	End year (decease, band dissolution...)	
	Tags	Foreign key to Tag entries	
Track	Title	Title of the track	artist_credit, medium, track, release, release_unknown_country, release_country, release_tag, l_release_release, artist_credit_name, artist, link, link_type
	Year	Year of the track's release	
	Month	Month of the track's release	
	Artist count	Number of artists involved	
	Tags	Foreign Key to Tag entries	
Tag	ID	MB unique identifier	tag
	Name	Name of tag	
Artist-Artist	–	–	l_artist_artist, link, link_type
Artist-Track	–	–	artist_credit, medium, track, release, release_unknown_country, release_country, release_tag, l_release_release, artist_credit_name, artist, link, link_type
Artist-Tag	–	–	artist_tag

**Fig. 2.** Pipeline to compose the MUSYNERGY recommender system.

Furthermore, Fig. 4 shows the confusion matrix with the co-location of some relevant music genres of the artists involved in a collaborative track. As we can see from the diagonal of such a matrix, rhythm-and-blues and hip-hop artists tend to mainly collaborate with artists with the same musical genre with co-location scores above 0.76. However, artists from other genres are more open to establish synergies with artists having different backgrounds. This is especially noticeable in indie musicians where 16.0% of their collaborations occur with hip-hop artists and 15.1% with musicians from the electronic genre. In this regard, it is important to remark we made use of all the music genres defined in the MBDB which had, at least, 5000 unique references to tracks for the development of MUSYNERGY. As a result, the framework considered 268 unique music genres.

The aforementioned results allowed us to establish that the recommender should consider dimensions related to age and time epoch of the artists along with the music genre of both songs and musicians. In the next section, we describe how we have integrated these findings in the HKG.

4.2. Definition of the heterogeneous knowledge graph

Based on the findings discussed in the preceding sections, we constructed an HKG to model the relationships among the key entities

considered in the study, namely *artists*, *tracks* and their *tags*. The structure of this graph is illustrated in Fig. 5. As we can see, each artist node comprises 5 different features where *begin date* indicates when their career started, *gender* indicates the artist's gender, *type* defines whether the artist is a solo performer, a band, or another category, *ended* is a boolean attribute that indicates whether such an artist is currently active in the music industry and *end date* indicates when that end occurred. Concerning the track nodes, their two attributes allow modeling their release date in the market. Last, the tag nodes incorporate a *label* attribute indicating the specific tag they represent.

The HKG also includes 9 distinct types of relations involving the aforementioned entities. This allows for a more varied palette of connections than previous studies whose HKGs comprised a smaller set of links [11].

In more in detail, the link *collab with* allows us to connect pairs of artists who collaborate on a track. As we will see in Section 4.3, this is the target link that MUSYNERGY aims to predict. Moreover, *personally related to* defines the personal or biographical relationships between artists (e.g. parent, sibling, romantic relationships and similar connections) as well as more indirect connections such as serving as the voice of another artist. Additionally, *musically related to* comprises musical relationships such as membership in a band, supporting musicians, performing under different stage names, musical leadership roles

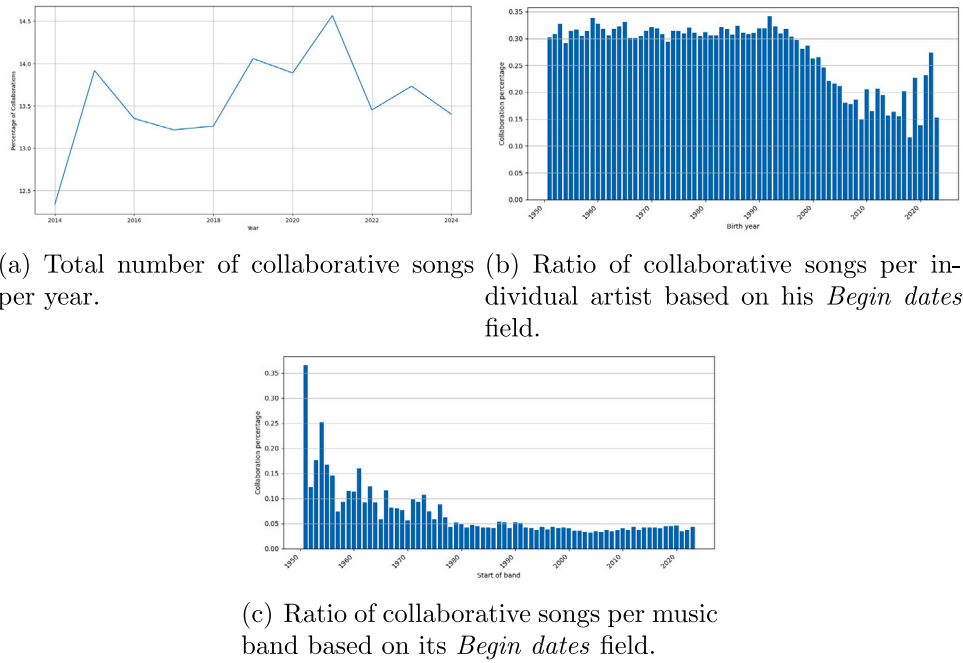


Fig. 3. Information about collaborative tracks in the MBDB.

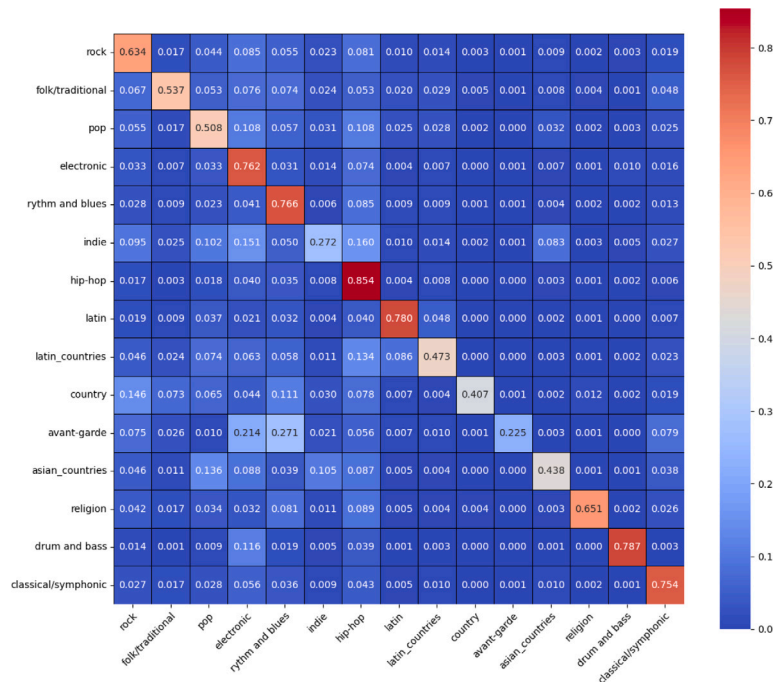


Fig. 4. Normalized confusion matrix with the primary music genre of the musicians involved in a collaborative track. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

such as conductors or artistic directors. Last, *linked to* relation includes the remaining connections that mostly represent organizational or mentorship ties, such as being a subgroup of another or having founded an artistic group or organization.

In addition, the links *worked in* and *worked by* define a bi-directional relationship between tracks and the artists who created them. Last, the relationships *has tag* and *tags* define connections between tracks and artists and their associated tags and labels.

It is important to note that modeling tags as standalone nodes rather than embedding them as attributes on track or artist nodes

enables a natural and efficient representation of many-to-many relationships. Each tag node exists uniquely within the graph and is connected to all associated tracks and artists, thereby avoiding the high-dimensional sparsity and preprocessing overhead inherent in one-hot or dense attribute encodings. This design also supports the definition of meaningful heterogeneous meta-paths, such as Artist→Tag→Artist or Track→Tag→Artist, which enrich node representations by explicitly propagating information along semantically relevant paths. Moreover, by treating tags as first-class graph entities, updates to tag vocabularies (e.g., additions, deletions, or reassignments) can be handled through

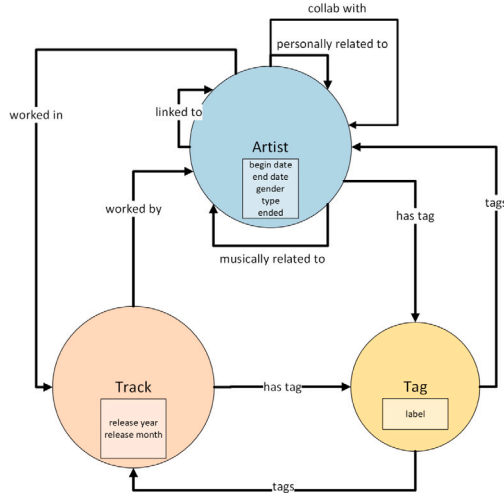


Fig. 5. Representation of the HKG capturing the relationships and attributes of artists, tracks, and genres. Colored rectangles denote the features associated with each artist and song node. The rectangle in each type of node indicates its associated attributes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

simple graph edits without the need to reengineer per-node feature spaces.

From a learning perspective, using tag nodes fosters parameter sharing and regularization across edge types, thereby reducing model complexity and improving generalization to novel tags or relationships. The resulting sparse, modular graph structure streamlines message-passing operations and facilitates the incorporation of tag-level semantics—such as embeddings or inter-tag connections—that further capture latent similarities among tags. Together, these advantages yield a flexible, scalable, and semantically enriched framework well suited to heterogeneous graph neural network architectures.

4.3. Problem formulation

At this point, we can formulate MUSYNERGY's recommendation task as an edge prediction problem: Given an HKG, $\mathcal{G}_{\mathcal{M}}^t$, following the schema described in Section 4.2, comprising a set of $\mathcal{M}$ musicians in a time interval t , find a mapping function $\mathcal{P}$ so that,

$$\mathcal{P}(\mathcal{G}_{\mathcal{M}}^t) \rightarrow \mathcal{C}_{\mathcal{M}}^A \quad (1)$$

where $\mathcal{C}_{\mathcal{M}}^A$ is the set of collaborations (*collab with* links) between musicians in $\mathcal{M}$ that occurred during the time interval $(t, t + \Delta]$.

4.4. Definition of the graph neural network architecture

In order to implement the function $\mathcal{P}$ described in the previous section, the internal architecture of MUSYNERGY's GNN is presented in Fig. 6. As shown, it takes as input a HKG $\mathcal{G}_{\mathcal{M}}^t$ which is processed by a four-layer pipeline where each layer is designed to address a specific stage of the processing pipeline while accounting for the heterogeneous characteristics of the input graphs.

The architecture consists of two successive layers of heterogeneous convolution operations, followed by a two-layer fully connected neural network to refine the artist node representations. In each convolutional layer, different types of relationships are handled using different convolutional operators to exploit the structural and semantic diversity of the graph,

On the one hand, the Graph Attention Convolution (GAT) [26] is applied to homogeneous relationships between artists (*collab with*, *linked to*, *musically related to* and *personally related to*). This layer leverages a

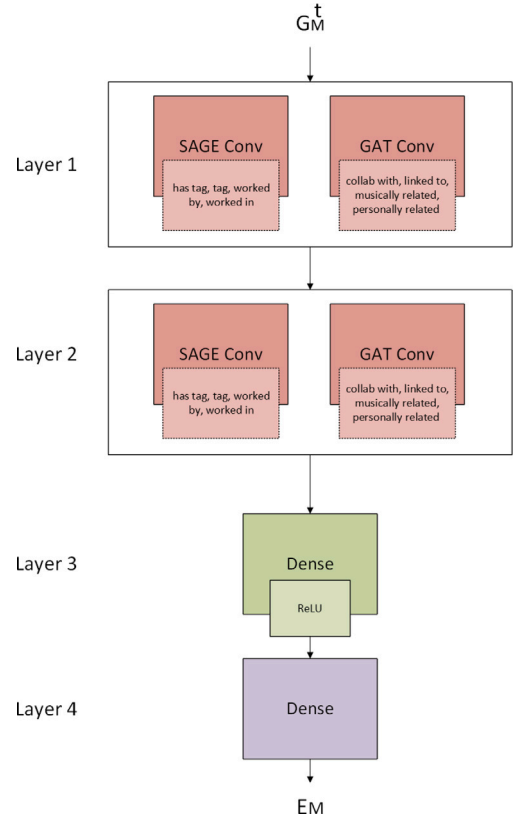


Fig. 6. Layer structure of MUSYNERGY.

multi-head attention mechanism to assign different levels of importance to a node's neighbors. As a result, the latent representation ε_v of a node v is computed using K attention heads, as shown in the following equation:

$$\varepsilon_v = \sigma \left(\frac{1}{K} \sum_{k=1}^K \sum_{u \in \mathcal{N}_v^k} \alpha_{vu}^k W^k \varepsilon_u \right) \quad (2)$$

Here, α_{vu}^k represents the attention coefficient that quantifies the contribution of neighbor u to node v in the k th attention head, and W^k denotes the corresponding weight matrix for the linear transformation applied in that head. This attention mechanism allows the model to dynamically weigh the importance of neighboring artist nodes, which is particularly suitable for capturing complex and varied interpersonal relationships among artists.

On the other hand, the GraphSAGE Convolution [27] is used for the other heterogeneous relationships between different types of nodes. This operator constructs node embeddings by progressively aggregating information from neighboring nodes. Specifically, the following equation describes how the representation of a node v , denoted as $h_v^{k,\zeta}$ at the k th layer and for a given edge type ζ , is computed using a mean aggregation strategy:

$$h_v^{k,\zeta} = \sigma \left(W_\zeta \cdot \text{MEAN} \left(\{h_v^{k-1,\zeta}\} \cup \{h_u^{k-1,\zeta} \mid u \in \mathcal{N}_v^{\zeta}\} \right) \right) \quad (3)$$

In this formulation, $\mathcal{N}_v^{\zeta}$ denotes the set of neighboring nodes of v connected via edges of type ζ , and W_ζ is the weight matrix associated with that specific edge type. The function σ represents a non-linear activation function applied to the aggregated result. GraphSAGE's ability to aggregate features from different node types is crucial for integrating information from tags and tracks, enriching the artist node features with contextual metadata.

Table 3
Model and training configuration for the experiments.

Type	Parameter	Value
Training	Loss	Binary Cross Entropy with logits (BCE)
	Train-test split	80% train, 20% test
	Learning rate	0,0001
	Weight decay	0.00001
	Optimizer	AdamW
	Batch size	128
	Neg. edge sampling rate	1.0
	Early Stopping	patience: 5
	Num. of epochs	100
SAGE (layers 1, 2)	Output size	64
GAT (layers 1, 2)	Num of heads	3
	Output size	64
Dense (layer 3)	Input size	128
	Output size	256
Dense (layer 4)	Input size	256
	Output size	64

This design leverages the strengths of both attention-based and aggregation-based convolutions, effectively capturing both the local and cross-domain relational patterns present in the HKG.

Furthermore, each convolutional layer aggregates neighborhood information using a mean aggregation strategy, balancing computational efficiency with representation richness. The output of the convolutional layer 2 is then processed by two fully connected layers (with a ReLU activation function in between) to project the features into the desired output space. Next, the output is L2-normalized to ensure that the learned artist embeddings have unit norm, promoting numerical stability and facilitating downstream tasks such as similarity computation.

In the final stage, this GNN architecture produces a matrix $\mathcal{E}_{\mathcal{M}}^A \in \mathbb{R}^{|\mathcal{M}| \times \rho}$, where each artist $m \in \mathcal{M}$ is represented by a vector with ρ dimensions. Based on this representation, we construct the matrix $\hat{\mathcal{C}}_{\mathcal{M}}^A \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{M}|}$ that assigns a score to each possible connection between pairs of artists. The computation is defined by the following equation,

$$\hat{\mathcal{C}}_{\mathcal{M}}^A = \text{sigmoid}(\mathcal{E}_{\mathcal{M}}^A \times (\mathcal{E}_{\mathcal{M}}^A)^T) \quad (4)$$

The sigmoid function converts the similarity scores into binary labels: for any pair of artists (u, v) , a value of $c_{uv} = 1$ in $\hat{\mathcal{C}}_{\mathcal{M}}^A$ indicates a recommendation that u and v should collaborate, whereas a value of 0 means no collaboration is suggested. This matrix serves as the foundation for the final output of the recommender system.

5. Evaluation of MUSYNERGY

In this section, we describe the evaluation strategy of our proposal and the main results and findings derived from it. The bottom row of the table indicates that the last dense layer produced embeddings consisting of 64 features. Consequently, the resulting node embedding matrix $\mathcal{E}_{\mathcal{M}}^A$ had dimensions of $|\mathcal{M}| \times 64$. Additionally, a negative edge sampling rate of 1.0 was applied during training. This implies that for every positive edge—representing a real collaboration between artists—a corresponding negative edge was artificially created to represent a non-existent collaboration. Thus, for each ground-truth matrix $\mathcal{C}_{\mathcal{M}}^A$, an equal number of artificial negative samples (labeled as 0) were incorporated alongside the true positive ones, enhancing the model's ability to distinguish between actual and potential collaborations.

5.1. Evaluation parameters

Table 3 shows the main configuration parameters of the GNN architecture and the training strategy followed in the study.

5.2. Evaluation metrics

Since the prediction task described in Section 4.3 can be viewed as a binary classification problem, we have used the F1 score, accuracy (ACC), and Area Under the ROC Curve (AUC) to evaluate the actual performance of our model.

It is important to note that the AUC metric is particularly useful when dealing with imbalanced datasets, as it reflects the model's ability to distinguish between the two classes regardless of the classification threshold. AUC values range from 0 to 1, with a value of 0.5 indicating random performance and values closer to 1.0 indicating better discriminative ability. This metric is widely used in link prediction tasks across graph-based models and has proven effective in evaluating classifier quality in sparse and noisy settings [28,29].

5.3. Baselines

In order to compare the suitability of our approach, we defined 4 different variations of the original MUSYNERGY architecture (Fig. 6), namely:

- 2 Convs+concat+2 Dense: GNN architecture like the one in Fig. 6 including a concatenation of the outputs from the convolutional layers 1 and 2 to feed layer 3.
- 4 Dense: FeedForward neural network comprising 4 stacked dense layers. This model receives as input the concatenated features of two different musicians and performs a binary classification to indicate whether they collaborate or not. Hence, the layers dimensions of this model are, $(s_i \times 256)$, (256×256) , (256×256) , (256×1) , where s_i is the size of the input vector comprising the features of the two target musicians.
- 1 Conv + 2 Dense: GNN architecture similar to the one in Fig. 6 but without the convolutional in layer 2.
- 1 Conv + 1 Dense: GNN architecture that like the one in Fig. 6 without the convolutional operator in layer 2 and the Dense layer 3.

Furthermore, we also include the PRESTO music recommender system as baseline [11]. Recall that this solution relied on a different GNN architecture with time-based operators which primarily relied on proprietary data feeds to compose the service.

5.4. Results analysis

Table 4 reports the performance of MUSYNERGY and the five aforementioned baselines based on the provided evaluation metrics. From this table, several interesting findings can be identified.

Table 4

Results of MUSYNERGY and the different baseline models. The values of Δ represent the prediction horizon in years. The best metrics for each Δ are shown in bold.

Model	Δ	ACC	Precision	Recall	F1-score	AUC
MUSYNERGY		0.9201	0.8756	0.9795	0.9246	0.9210
4 Dense		0.7834	0.8727	0.6636	0.7539	0.8048
2 Convs + concat + 2 Dense		0.9166	0.8706	0.9788	0.9215	0.9127
1 Conv + 2 Dense	1	0.8722	0.8391	0.9210	0.8782	0.8873
1 Conv + 1 Dense		0.8671	0.8300	0.9234	0.8742	0.8860
PRESTO		0.7150	–	–	0.7650	–
MUSYNERGY		0.9189	0.8694	0.9859	0.9240	0.9166
4 Dense		0.7155	0.8575	0.5169	0.6450	0.7253
2 Convs + concat + 2 Dense		0.9154	0.8657	0.9833	0.9208	0.9131
1 Conv + 2 Dense	3	0.8817	0.8310	0.9583	0.8901	0.8919
1 Conv + 1 Dense		0.8523	0.8159	0.9100	0.8604	0.8722
MUSYNERGY		0.9131	0.8604	0.9862	0.9190	0.9137
4 Dense		0.6744	0.8493	0.4241	0.5657	0.6790
2 Convs + concat + 2 Dense		0.9129	0.8591	0.9878	0.9189	0.9116
1 Conv + 2 Dense	5	0.8739	0.8267	0.9462	0.8824	0.8915
1 Conv + 1 Dense		0.8689	0.8199	0.9454	0.8782	0.8873

First, across all prediction horizons, MUSYNERGY achieves the highest F1-score and AUC, demonstrating both strong discriminative power and a balanced precision–recall trade-off. For $\Delta = 1$ year, MUSYNERGY obtains an AUC of 0.9201 and F1=0.9246, outperforming the nearest competitor (2 Convs+concat+2 Dense) by approximately 0.008 in AUC and 0.003 in F1. This margin persists across longer horizons ($\Delta = 3, 5$ years), indicating that its heterogeneous GAT+GraphSAGE convolutional architecture yields robust link embeddings even as temporal gaps widen. Importantly, MUSYNERGY also clearly outperforms the PRESTO baseline in terms of both accuracy and F1 score.

Unlike the purely dense model (4 Dense), which suffers from both low recall (dropping to 0.4241 at $\Delta = 5$) and reduced accuracy (falling to 0.6744 at $\Delta = 5$), MUSYNERGY maintains exceptionally high Recall (~ 0.98 across all horizons), strong Precision (~ 0.876 at $\Delta = 1$, decreasing slightly to ~ 0.860 at $\Delta = 5$) and accuracy around 0.92 (0.9201 at $\Delta = 1$, 0.9131 at $\Delta = 5$). This reflects its superior capability to recover true future links while minimizing false positives, underscoring the critical importance of graph-based neighborhood aggregation over purely feature-based classifiers.

It was also interesting to observe that, removing the second convolutional block (1 Conv+2 Dense) or the final dense layer (1 Conv+1 Dense) degrades both F1 and AUC by 3%–5%, confirming that the two-stage message-passing (GAT for artist–artist links and GraphSAGE for cross-type edges) is critical to capture the heterogeneous semantic paths in the HKG. The variant that concatenates both convolutional outputs (2 Convs+concat+2 Dense) performs almost on par with MUSYNERGY, but still trails slightly, suggesting that the learned linear projection, rather than naïve concatenation, better fuses multi-relation signals.

Finally, as the prediction window expands from 1 to 5 years, all models experience mild performance decay. MUSYNERGY’s AUC drops only modestly by 0.0073, whereas the baselines degrade by up to 0.13. This resilience highlights MUSYNERGY’s capacity to model long-term latent collaboration patterns through its unified embedding space and edge-scoring via Eq. (4).

In summary, the full MUSYNERGY architecture, combining GAT-based attention over homogeneous artist relations and GraphSAGE aggregation over heterogeneous edges, followed by a two-layer MLP and L2-normalization consistently outperforms simpler or ablated variants. It achieves the best balance between correctly identifying future collaborations (Recall), avoiding spurious links (Precision), and maintaining high overall discrimination (AUC and F1), thereby demonstrating its effectiveness for the music collaboration recommendation task.

Table 5

Comparison between initial version of MUSYNERGY solely fed with the MBDB (MUSYNERGY_{mb}) and its enrichment with Last.fm data (MUSYNERGY_{mb,lfm}). The values of Δ are defined in years. The best value for each (metric, Δ) pair is shown in bold.

Model	Δ	ACC	Precision	Recall	F1-score	AUC
MUSYNERGY _{mb,lfm}	1	0.9329	0.9107	0.9599	0.9346	0.9646
MUSYNERGY _{mb}		0.9201	0.8756	0.9795	0.9246	0.9210
MUSYNERGY _{mb,lfm}	3	0.9321	0.9016	0.9700	0.9345	0.9599
MUSYNERGY _{mb}		0.9189	0.8694	0.9859	0.9240	0.9166
MUSYNERGY _{mb,lfm}	5	0.9281	0.8998	0.9635	0.9306	0.9619
MUSYNERGY _{mb}		0.9131	0.8604	0.9862	0.9190	0.9137

5.5. Enrichment with Last.fm data

As a final step in evaluating our approach, we studied the impact of enhancing MUSYNERGY with a proprietary data feed. In particular, we utilized the Last.fm (LFM) API.² Last.fm is a well-known online music service that offers an extensive API from which which provides numerous endpoints that yield information about artists and tracks. We merged this feed with MBDB in order to compose a new input dataset. In this context, we leverage the features provided by the Last.fm API to enrich the HKG described in Section 4.2 in two aspects,

- We added a *popularity* score to the *artist* and *track* nodes to measure the public appeal within the music industry.
- We included a *last-fm match* link between artists which encodes the similarity score between pairs of musicians according to the Last.fm service.

Table 5 shows a comparison of the results obtained by the model when it was solely fed with the MBDB (MUSYNERGY_{mb}) and the one enriched with Last.fm data (MUSYNERGY_{mb,lfm}).

From the table, it is evident that MUSYNERGY_{mb,lfm} consistently outperformed MUSYNERGY_{mb} in most evaluation metrics. Specifically, it achieved the highest ACC across all time horizons, indicating better overall classification capability. Similarly, its precision, which measures the correctness of positive predictions, was also superior, indicating that MUSYNERGY_{mb,lfm} produced fewer false positives. Furthermore, the F1-score, which balances precision and recall, and the AUC, which evaluates the model’s ability to distinguish between classes across thresholds, were both consistently higher in MUSYNERGY_{mb,lfm}, highlighting the enhanced robustness and generalization capability provided by the LFM dataset.

Despite these advantages, MUSYNERGY_{mb} achieved the highest recall in all settings. For example, in the 1-year horizon, MUSYNERGY_{mb} reached a recall of 0.9795, compared to 0.9599 in its enhanced version. This trend continues for 3-year and 5-year horizons, with MUSYNERGY_{mb} achieving 0.9859 and 0.9862, respectively, both values notably higher than those of MUSYNERGY_{mb,lfm} (0.9700 and 0.9635). This observation suggests that the model relying solely on open data was more aggressive in identifying positive instances, possibly at the cost of increased false positives, as reflected in its lower precision and overall accuracy. The higher recall means that the model is less likely to miss positive cases. In contrast, MUSYNERGY_{mb,lfm} adopted a more balanced strategy, resulting in improved precision and F1-score, but slightly reduced recall.

Given the artists’ popularity extracted from LFM, we also conducted a new experiment to assess the suitability of MUSYNERGY based on this attribute. To do so, we stratified the dataset fusing the MBDB and LFM feeds by applying different percentile-based popularity thresholds. This way, we trained a MUSYNERGY instance using only the top 10% most

² <https://www.last.fm/api>.

Table 6

Results of MUSYNERGY_{mb,lfm} based on the selected popularity quantile of musicians and the time horizon Δ (in years). The best value for each (metric, Δ) tuple is shown in bold.

Δ	Pop. Quantile	ACC	Precision	Recall	F1-score	AUC
1	100%	0.9329	0.9107	0.9599	0.9346	0.9646
	50%	0.9473	0.9178	0.9826	0.9491	0.9410
	25%	0.9317	0.8813	0.9978	0.9360	0.8871
	10%	0.9196	0.8617	0.9996	0.9255	0.8500
3	100%	0.9321	0.9016	0.9700	0.9345	0.9599
	50%	0.9483	0.9211	0.9806	0.9499	0.9409
	25%	0.9325	0.8866	0.9919	0.9363	0.8888
	10%	0.9198	0.8621	0.9996	0.9258	0.8481
5	100%	0.9281	0.8998	0.9635	0.9306	0.9619
	50%	0.9462	0.9167	0.9816	0.9481	0.9408
	25%	0.9321	0.8849	0.9934	0.9360	0.8891
	10%	0.9188	0.8610	0.9989	0.9248	0.8485

popular musicians in the dataset, another instance using the top 25%, and another using the top 50% of the most popular artists. The results of this evaluation are shown in Table 6.

These results demonstrate a strong dependency of predictive performance on the *popularity quantile* used to train the model. At a 10% rate, the model achieved near-perfect recall (up to 0.9996 at $\Delta=1$), indicating that nearly all true future collaborations among these superstars were detected. However, precision and AUC suffered (e.g. AUC ~ 0.85), reflecting an inflated false-positive rate when the graph contained only high-degree nodes with heavily overlapping connection patterns.

As the quantile expanded to 25% and 50%, introducing progressively less well-known artists, precision and overall accuracy improved markedly. At 50%, the model attained its highest accuracy (ACC = 0.9473 at $\Delta = 1$ year) and balanced F1-score (F1 = 0.9491), suggesting that including mid-tail artists yields a more stable balance between positive and negative examples. This intermediate coverage appears to mitigate the extreme homogeneity of the head-only graph (10%) while avoiding the noise introduced by the long tail.

When trained on the full artist set (100% rate), which incorporated both super-stars and the long-tail of niche or emerging artists, the model achieved maximal discrimination (AUC up to 0.9646), demonstrating its ability to distinguish true from spurious edges across a highly heterogeneous degree distribution. Nonetheless, accuracy and precision declined slightly relative to the 50% subset, as the sparsely connected fringe of obscure artists introduces additional class-imbalance challenges and noisier feature representations.

Regarding the temporal horizon Δ exerted only a modest effect on these trends: the optimal rate-dependent trade-off remained consistent, with the 50% subset excelling in accuracy and F1-score, and the 100% set excelling in AUC.

In summary, a careful selection of the artists' target popularity would allow practitioners to tailor the model's behavior—prioritizing recall among top-tier collaborators at low rates, maximizing overall accuracy at intermediate rates, or achieving maximal discrimination across the entire popularity spectrum at full coverage.

5.6. Execution performance

Finally, we evaluated the time required by MUSYNERGY to perform its recommendation task. In this sense, Fig. 7 shows the distribution of CPU time required by the framework and its baselines based on 10,000 executions.

As we can see from this figure, our recommendation system required, on average, about 9 ms to generate a recommendation. The execution times required by the other baselines are consistent with the complexity of the architectures.

6. Illustrative examples

For completeness, Table 7 presents some examples of collaborations predicted by MUSYNERGY for different time horizons Δ . As we can see, our solution was able to anticipate synergies between very popular artists such as the band Coldplay and Lady Gaga, who come from different musical backgrounds.

Notably, MUSYNERGY successfully predicted the rare collaboration between Hans Zimmer, a renowned Hollywood composer, and Lady Gaga, a leading pop singer, one year in advance. Furthermore, another noteworthy prediction made by the framework was the collaboration between Madonna, one of the most iconic pop artists of all time at age 66, and the 35-year-old artist The Weeknd, for the song *Popstar*, two years before it occurred. Besides, our framework also predicted collaborations between less popular artists like the one involving Noah Cyrus, and American singer and actress and Jimmie Allen, an American country music songwriter, 5 years before it occurred.

7. Conclusions and future work

The music industry has emerged as a highly promising domain for the development of intelligent solutions based on artificial intelligence and data mining techniques. In this context, one of the most prominent research directions has been the design of recommendation systems aimed at assisting listeners in the discovery of new and relevant songs. Nevertheless, there remains a notable lack of recommender systems tailored to the needs and interests of music artists. Moreover, most existing approaches depend on data sources that are either directly or indirectly derived from proprietary music streaming platforms, which imposes significant restrictions on data accessibility and limits the applicability and reproducibility of such systems.

In this work, we introduced MUSYNERGY, a novel recommender system designed to assist musicians in identifying other artists whose musical style and background make them suitable collaborators for joint creative projects. MUSYNERGY exclusively relies on MusicBrainz, an open and collaboratively maintained database, which we model as a heterogeneous knowledge graph (HKG). The evaluation results demonstrated that the system achieved F1 scores close to 0.92, clearly outperforming comparable solutions that rely on proprietary datasets.

In future work, we aim to enhance the HKG by incorporating additional features from other open data sources, such as sentiment analysis and language detection from lyrics, as well as metadata related to the artists (e.g., the number of incoming links to their Wikipedia pages). This enrichment is expected to enable more accurate and context-aware recommendations by accounting for a broader range of relevant attributes.

Code availability

The source code of this project is available at <https://github.com/aleferu/musicbrainz>.

CRedit authorship contribution statement

Alejandro Fernandez-Sanchez: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pedro J. Navarro:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Fernando Terroso-Saenz:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors have declared no conflict of interest

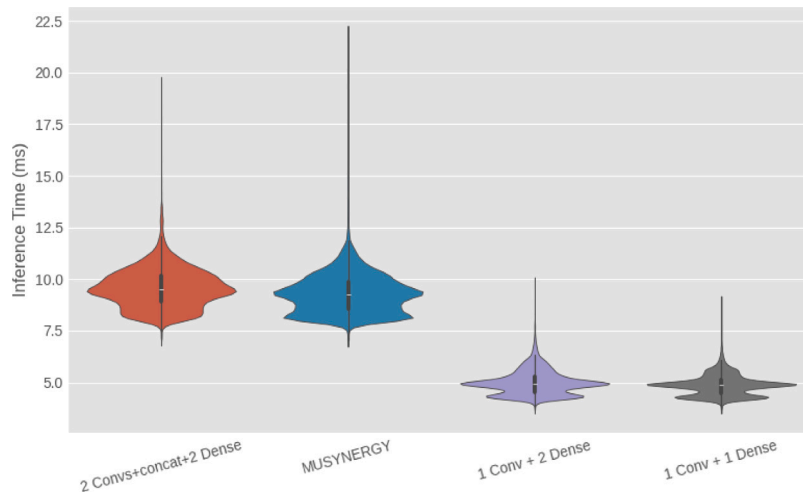


Fig. 7. Distribution of the CPU time of MUSYNERGY and its three variations.

Table 7

Examples of musical collaborations suggested by MUSYNERGY for different time horizons Δ .

Δ	Artist A	Primary genre A	Artist B	Primary genre B	Track title	Track genres
1	Coldplay	Rock	TINI	Pop	WE PRAY	Pop, Rock, Hip-Hop
	Calvin Harris	Electronic	Ellie Goulding	Pop	Free	Pop, Rock, Hip-Hop, Electronic, Rhythm and Blues
	Andrea Bocelli	Classical and OST	Lauren Daigle	Religion	Canto Della Terra	Pop, Latin Countries, Classical and OST
3	Hans Zimmer	Classical and OST	Lady Gaga	Pop	Hold My Hand	Pop
	Madonna	Pop	The Weeknd	Rhythm and Blues	Popular	Electronic, Rock, Rhythm and Blues, Sad, Hip-Hop, Pop
	Eve	Rock	Tatsuya Kitani	Asian countries	Love Song	Asian countries, Rock, Pop
5	Mariah Carey	Rhythm and Blues	Ariana Grande	Pop	Oh Santa!	Pop
	Lady Gaga	Pop	Elton John	Rock	Sine From Above	Electronic, Pop
	Noah Cyrus	English countries	Jimmie Allen	Country	This is us	English countries, Country

Data availability

The data is open source and is available.

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